

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester III Examination, 2024–25

Subject: MATHEMATICS
Paper No. MTB-302: Differential Equations

Time: 3 Hours

Full Marks: 70

Note: Answer five questions including Question No. 1 which is compulsory. The figures in the right-hand margin indicate the split mark distributions for individual subsections.

1. Answer all parts of this question (Compulsory):

- (a) Find the differential equation of the family of circles whose y -axis acts as the tangent at the origin. [2]
- (b) Find the explicit numerical value of the constant m for which the function $y = (A+Bx)2^x + Ce^x$ is a verified solution of the homogeneous linear differential equation:

$$\frac{d^3y}{dx^3} + m \frac{d^2y}{dx^2} + 4y = 0$$

[2]

- (c) Find an exact integrating factor (I.F.) for the non-exact differential equation:

$$(x^3e^x - my^2)dx + mxy dy = 0$$

[2]

- (d) Solve the non-linear first-order differential equation:

$$\frac{dy}{dx} = (4x + y + 1)^2$$

[2]

- (e) Let $y(x)$ be the solution of the boundary value constraints $x^2 \frac{dy}{dx} = y + 1$ with the initial value condition $y(1) = 0$. Evaluate:

$$\lim_{x \rightarrow \infty} y(x)$$

[2]

- (f) Find the general primitive solution of the non-linear differential equation containing squared derivative forms:

$$x \left(\frac{dy}{dx} \right)^2 - y \left(\frac{dy}{dx} \right) + a = 0$$

[2]

- (g) Show that the infinite series expansion containing the standard Legendre Polynomials satisfies the identity:

$$\sum_{n=0}^{\infty} P_n(x) = \frac{1}{\sqrt{2-2x}}$$

[2]

2. (a) Solve the first-order linear differential equation:

$$\sin x \frac{dy}{dx} + 3y = \cos x$$

[6]

- (b) Find the general primitive solution of the differential equation:

$$(y^2 + 2x^2y)dx + (2x^3 - xy)dy = 0$$

[6]

3. (a) Find both the general solution and the singular solution of the first-order higher-degree differential equation:

$$\left(\frac{dy}{dx}\right)^3 - 4xy\left(\frac{dy}{dx}\right) + 8y^2 = 0$$

[6]

- (b) Solve the fourth-order non-homogeneous linear differential equation:

$$\frac{d^4y}{dx^4} + \frac{d^2y}{dx^2} + y = 2e^{-x} \sin 2x$$

[6]

4. (a) Test the mathematical exactness of the second-order linear ordinary differential equation:

$$(1 + x^2)\frac{d^2y}{dx^2} + 4x\frac{dy}{dx} + 2y = \sec^2 x$$

and hence obtain its general solution.

[6]

- (b) By reducing the equation to its standard normal form, solve:

$$\frac{d^2y}{dx^2} - 2 \tan x \frac{dy}{dx} + 5y = e^x \sec x$$

[6]

5. Solve the variable-coefficient second-order ordinary differential equation in power series form around the regular singular point:

$$x(1-x)\frac{d^2y}{dx^2} - (1+3x)\frac{dy}{dx} - y = 0$$

[12]

External Portals & Academic Resources:

Examination Portal: <https://bhu-exam.vercel.app/> — Student Resource Center: <https://quashan-me.vercel.app/>
Academic Portfolio: <https://me-aryan.vercel.app>